

BCA 3rd SEMESTER EXAM. 2014
SOFTWARE ENGINEERING CODE- 303304

Time: 3 hours

Full Marks: 60

Instructions:

- i. The Marks are indicated in the right -hand margin.
- ii. There are **SEVEN** questions in this paper.
- iii. Attempts **FIVE** question in all.
- iv. Question Nos. **1** and **2** are compulsory.

1. Answer the following as directed (any six).

6*2=12

(a) What is the main concern of software engineering area?

- | | |
|-----------------------------|---------------------------|
| (I) Hardware configuration | (II) Software Production |
| (III) Network Configuration | (IV) Software Reusability |

(b) The worst type of coupling is

- | | |
|----------------------|-----------------------|
| (I) Data coupling | (II) Control coupling |
| (III) Stamp coupling | (IV) Content coupling |

(c) All the modules of the system are integrated and tested as complete system

In the case of

- | | |
|------------------------|------------------------|
| (I) Bottom-up testing | (II) Bing-band testing |
| (III) top-down testing | (IV) Sandwich testing |

(d) SRS in also known as specification of

- | | |
|-------------------------|-------------------------|
| (I) White-box testing | (II) Stress testing |
| (III) Black-box testing | (IV) Integrated testing |

(e) COCOMO stands for _____.

(f) Abstract data type does not allow inheritance.

- | | |
|----------|------------|
| (I) True | (II) False |
|----------|------------|

(g) The too that supports different stage of software development life cycle are called

- | | |
|-----------------|----------------|
| (I) CASE tool | (II) CAME tool |
| (III) CAQE tool | (IV) CARE tool |

(h) Change the made to the system to reduce the future system failure change is called ____.

(i) Requirement can be refined using

(I) Waterfall model

(II) Prototyping model

(III) Evolutionary model

(IV) Spiral model

(j) The main purpose of integration testing is to find

(I) Design errors

(II) Analysis errors

(III) Procedure errors

(IV) Interface errors

2. Answer any three of the following:

3*4=12

(a) What is the prototype? Under what circumstances it is beneficial to construct

Prototype?

(b) List of the important shortcoming of LOC for use as a software size metric.

(c) What do you mean by balancing a DFD? Illustrate with example.

(d) What is the difference between Black -box and white-box testing?

(e) How can you determine the number of latent defects in a software product during the testing phase?

Answer any three of the following:

3*12=36

3. (a) During the software testing process why is the reliability growth initially high but

Shows down later on?

6

(b) What is the six-sigma quality initiative? Explain.

6

4. (a) What do you mean by side effects of a function call? Give example.

6

(b) What is regression testing? When is it done? Why it is necessary? How is it performed? 6

5. (a) Distinguish between DFD and flow –chart.

6 (b) What is the main shortcoming of DFD as a tool for performing structured analysis?

6

6. (a) What do you understand by siding window planning? Give example and its advantage. 6

(b) What are the different categories of software development projects according to the

COCOMO estimation model? Give example for each.

6

7. (a) How are risks associated with a project handed in the spiral mode of software

development?

6

(b) Why is it not prudent to use the iterative waterfall model for developing very large

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BCA 3rd Semester Theory Examination, 2017

Software Engineering

3 Hrs

Full Marks : 60

Instructions :

The questions are of equal value.

- There are Seven questions in this Paper.*
- Attempt Five questions in all.*
- Question Nos. 1 & 2 are compulsory.*

Answer any six questions of the following: 2×6=12

What is Black Box Testing?

What is Exhaustive Testing?

What is Static Analysis?

Explain the design steps of the transform mapping.

How is effort measured?

State the objectives and guidelines for debugging.

Describe the characteristics of Software.

Distinguish between verification and validation.

What do you mean by staffing?

What is Bottom-Up Design?

P.T.O.

2. Answer any three of the following:

- (a) Discuss the Different Levels of Testing.
- (b) Spiral model in Software Engineering.
- (c) What are Test Oracles?
- (d) Write a Note on Software Design Phases.
- (e) What are the different methods used for Monitoring Project?

Answer any three questions.

- 3. What is Feasibility Study? Describe Types of Feasibility Study. Explain Feasibility Study Process.
- 4. What is Structural Testing? Explain any Two Techniques in it.
- 5. What is Process Model? Explain Waterfall Model and its limitations.
- 6. What are the different approaches used for the Verification of a Design Document.
- 7. What is build and fix model or ad hoc model? Also explain its Advantages and Disadvantages.

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BCA 3rd Semester Exam., 2018

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer of the following (any six) : $2 \times 6 = 12$

- (a) SDLC stands for
 - (i) Software Development Life Cycle
 - (ii) System Development Life Cycle
 - (iii) Software Design Life Cycle
 - (iv) System Design Life Cycle
- (b) Which one of the following is a functional requirement?
 - (i) Maintainability
 - (ii) Portability
 - (iii) Robustness
 - (iv) None of the above

(Turn Over)

(2)

- (c) Which of the following is an indirect measure of product?
 - (i) Quality
 - (ii) Complexity
 - (iii) Reliability
 - (iv) All of the above
- (d) In size-oriented metrics, metrics are developed based on the
 - (i) number of functions
 - (ii) number of user inputs
 - (iii) number of lines of code
 - (iv) amount of memory usage
- (e) Which is the first step in the software development life cycle?
 - (i) Analysis
 - (ii) Design
 - (iii) Problem/Opportunity identification
 - (iv) Development and documentation
- (f) Which tool is used for structured designing?
 - (i) Program flowchart
 - (ii) Structure chart
 - (iii) Dataflow diagram
 - (iv) Module

AK9/290

(Contd.)

- (g) What is the final outcome of requirement analysis and specification phase?
- (i) The SRS document
 - (ii) Code
 - (iii) User manual
- (h) DFD shows
- (i) the flow of data
 - (ii) the processes
 - (iii) the areas where they are stored
 - (iv) All of the above
- (i) Which box specifies the behaviour of a system or a part of a system?
- (i) State box
 - (ii) Clear box
 - (iii) Black box
 - (iv) None of the above
- (j) The spiral model has two-dimensions, namely ____ and ____.
- (i) diagonal, angular
 - (ii) radial, perpendicular
 - (iii) radial, angular
 - (iv) diagonal, perpendicular

2. Answer any three of the following questions:

4x3

- (a) What are the standards to measure software quality?
 - (b) How to estimate size of a project?
 - (c) What is the need of functional independence?
 - (d) What are the differences between function-oriented and object-oriented design?
 - (e) Explain SEI CMM levels.
3. Explain classical waterfall model with activities undertaken during each phase.
4. Explain in detail integration test approaches.
5. Which symbols are used for designing DFDs? Explain with example DFD. Explain synchronous and asynchronous dataflow with example.
6. Elaborate in detail parts of SRS document with example.
7. Discuss in detail software maintenance process models.

Code : 303304

BCA 3rd Semester Exam., 2019

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Choose the correct answer (any six) : $2 \times 6 = 12$

- (a) The context diagram of a DFD is also known as
 - (i) level 0 DFD
 - (ii) level 1 DFD
 - (iii) level 2 DFD
 - (iv) None of the above
- (b) The black box is
 - (i) invoking the system in terms of input and output
 - (ii) independent
 - (iii) assuming input and output is stable always
 - (iv) None of the above

20AK/412

(Turn Over)

(2)

- (c) The primary aim of software engineering is to provide
 - (i) reliable software
 - (ii) according to requirement complete software
 - (iii) cost-effective software
 - (iv) All of the above
- (d) Which of the following is not characteristic of SRS document?
 - (i) Ambiguous
 - (ii) Traceable
 - (iii) Concise
 - (iv) Verifiable
- (e) The aggregation relationship cannot be reflexive and symmetric but is transitive
 - (i) False
 - (ii) True
 - (iii) None of the above
- (f) The worst type of coupling is
 - (i) control
 - (ii) data
 - (iii) content
 - (iv) stamp

20AK/412

- (g) In the context of use case diagram, the stick person icon is used to represent
- (i) internal system
 - (ii) external system
 - (iii) human user
 - (iv) None of the above
- (h) Which is the most important feature of spiral model?
- (i) Risk management
 - (ii) Efficiency management
 - (iii) Quality management
 - (iv) Performance management
- (i) If every requirement can be checked by cost-effective process, then SRS is
- (i) traceable
 - (ii) modifiable
 - (iii) complete
 - (iv) verifiable
- (j) Waterfall model not suitable for
- (i) small projects
 - (ii) complex projects
 - (iii) accommodating changes
 - (iv) diagonal, perpendicular

(Turn Over)

2. Answer any three of the following :
- (a) What is the difference between black box and white-box testing?
 - (b) What is software? What are key fundamental software engineering activities and general issues?
 - (c) List and explain any five software engineering code of ethics.
 - (d) What are the differences between function-oriented and object-oriented designs?
 - (e) Discuss prototyping model. What is the effect of designing a prototype on overall cost of software project.
3. Explain rapid application development phase.
4. Explain software maintenance process models.
5. Identify the requirements of algebraic specification in order to define a system.
6. Explain in detail the structure of requirements document according to IEEE.
7. What is project scheduling? Explain with example.

Code : 303304

BCA 3rd Semester Exam., 2020

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. 1 and 2 are compulsory.

1. Answer any six questions : 2×6=12

- (a) What is Software Quality Assurance?
- (b) What is Software Testing?
- (c) What are verification and validation?
- (d) What is software crisis?
- (e) Are lines-of-code (LOC) a useful productivity measure?
- (f) Explain data dictionary.

AK-21/393

(Turn Over)

(2)

(g) Describe the characteristics of good SRS.

(h) Draw a block diagram of software life cycle model.

(i) What is software maintenance and categories of maintenance?

(j) What is software risk?

2. Answer any three of the following : 4×3=12

- (a) Describe the characteristics of software.
- (b) Explain the software requirement analysis and modeling.
- (c) Narrate the importance of software specification of requirements.
- (d) What are the challenges in software?
- (e) What are test oracles?

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3. What is transform mapping? Explain the design steps of the transform mapping. 12

4. Discuss how the testing models may be used together to test a program schedule. 12

AK-21/393

(Continued)

5. Explain the need for software measures and describe the process metrics and product metrics. 12
6. Compare the incremental model and the spiral model. 12
7. What is process model? Explain waterfall model along with its limitations. 12

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Code : 303304

BCA 3rd Semester Exam., 2022

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. 1 and 2 are compulsory.*

1. Answer any six of the following : **2×6=12**

- (a) What is the goal of software engineering?*
- (b) What are the two types of software products?*
- (c) What is system engineering?*
- (d) What is the basic difference between process and method?*
- (e) What is software prototyping?*

- (f) Define behaviour modeling.
- (g) How do you evaluate user interface?
- (h) What is the software architecture?
- (i) What is stress testing?
- (j) Why does the software need maintenance?

2. Answer any *three* of the following : 4×3=12

- (a) Briefly explain what is Delphi cost estimation technique.
- (b) Explain boundary value analysis.
- (c) Write a short note on user interface design process. <https://www.akubihar.com>
- (d) Write a short note on data modeling.
- (e) Identify the umbrella activities in software engineering process

3. Explain how both waterfall model and prototyping model can be accommodated in the spiral process model. 12

4. Describe two metrics which are used to measure the software. Discuss clearly the advantages and disadvantages of these metrics. 12

5. Describe the concepts of cohesion and coupling. State the difference between cohesion and coupling with suitable example. 12
6. Explain black box testing methods and its advantages and disadvantages. 12
7. With respect to software quality assurance, discuss the estimation of maintenance costs. 12

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BCA 3rd Semester Exam., 2024

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.*
- (ii) There are **SEVEN** questions in this paper.*
- (iii) Attempt **FIVE** questions in all.*
- (iv) Question Nos. 1 and 2 are compulsory.*

1. Answer any six of the following questions :

2×6=12

- (a) What is Waterfall Model?*
- (b) Name a popular version control system used in software development.*
- (c) What is the purpose of software testing?*
- (d) What is an agile development methodology?*
- (e) Define software architecture in one sentence.*

- (f) What is the role of a software requirement specification?
- (g) Name a common type of software bug.
- (h) What is the concept of ISO 9000?
- (i) What is the benefit of using design patterns?

2. Answer any *three* of the following : 4×3=12

- (a) Define the term 'software requirements' in the context of software engineering.
- (b) Explain the difference between ISO 9000 and SEI CMM quality assurance.
- (c) How does object-oriented design improve software maintainability and reusability compared to procedural programming?
- (d) What is the significance of reverse engineering?
- (e) Discuss the concept of estimation of maintenance costs.

3. Explain the Software Development Life Cycle (SDLC). Describe the typical phases involved in a traditional SDLC model (like Waterfall). 12

4. Define software requirements engineering. Discuss the difference between functional and non-functional requirements, providing examples of each. 12
5. Describe the concept of software design. Explain the difference between high-level and low-level design. 12
6. Explain the importance of software testing in the software development process. Describe the different levels of software testing and the purpose of each. 12
7. Discuss the concept of software quality assurance. Discuss one of the popular maintenance process models. 12

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Code : 303304

BCA 3rd Semester Exam., 2025

SOFTWARE ENGINEERING

Time : 3 hours

Full Marks : 60

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **SEVEN** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question Nos. **1** and **2** are compulsory.

1. Answer any six of the following questions :
2×6=12

- (a) What is ISO 9000?
- (b) Name a common traditional Software Development Life Cycle (SDLC) model.
- (c) What are the two main categories of software requirements?
- (d) What is COCOMO?
- (e) What is the term for breaking down a software system into smaller, manageable parts?

26AK/463

(Turn Over)

(2)

- (f) What is DFD?
- (g) What is the purpose of version control in software development?
- (h) What is the process of fixing defects in a released software product called?
- (i) What is object-oriented design?
- (j) What is the practice of continuously integrating code changes into a shared repository called?

2. Answer any three of the following questions :
4×3=12

- (a) Define the term 'software requirements' in the context of software engineering.
- (b) Explain the difference between ISO 9000 and SEI CMM Quality Assurance.
- (c) How does object-oriented design improve software maintainability and reusability compared to procedural programming?
- (d) What is the significance of reverse engineering?
- (e) Discuss the concept of estimation of maintenance costs.

26AK/463

(Continued)

3. Explain the Software Development Life Cycle (SDLC). Describe the typical phases involved in a traditional SDLC model. 12
4. Define software requirements engineering. Discuss the difference between functional and non-functional requirements providing examples of each. 12
5. Describe the concept of software design. Explain the difference between high-level and low-level designs. <https://www.akubihar.com> 12
6. Explain the importance of software testing in the software development process. Describe the different levels of software testing and the purpose of each. 12
7. Discuss the concept of Software Quality Assurance. Discuss one of the popular maintenance process models. 12

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